

Aarti Darji

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EDUCATION

M.S. Computer Science, *New York University, Courant Institute*

Jan 2026-Dec 2027

B.S. Data Science & Psychology, *University of Texas at Arlington*

Aug 2021-May 2025

SKILLS

Programming Languages	Python, R, SAS, MATLAB, SQL, JavaScript, C++, HTML/CSS
Libraries/Frameworks	React.js, Node.js, Flask, Pandas, Matplotlib, Seaborn, TensorFlow, Pytorch
Lightning, CUDA, Scanpy,	OpenCV, ggplot, tidyverse.
Tools	Git, Github, Unix/Linux, Bash, Slurm, ImageJ, QuPath, Figma, Microsoft Office, FSL, FreeSurfer.
Relevant Coursework	Programming Languages, Operating Systems, Data Mining, NLP, Machine Learning, Statistical Inference, Behavioral Genetics.

RESEARCH EXPERIENCE

Graduate Research Assistant - SSL for Mammography Imaging

Jan 2026 - Present

Advisor: Dr. Yiqiu Shen

New York University

- Fine-tuned Mammogram foundation models NYU mammograms with breast-level cancer labels using custom HDF5 dataloader, stratified sampling, and data augmentations.
- Conducted hyperparameter sweep across learning rates, sampling ratios and batch size configurations for pre-trained foundational models.

Research Assistant I - Analyzing Gulf War Illness in Veterans

June 2025- Dec 2025

Advisor: Dr. Deborah Little

UTHealth Houston

- Curated and maintained raw and processed MRI data for Gulf War Illness studies in veterans.
- Performed diffusion MRI analysis using FSL and FreeSurfer for structural and tract-based insights.
- Quantitatively rated 100+ MRI scans using Anterior Atrophy and Fazekas white matter hyperintensity scores.

Undergraduate Research Intern - LLM Agent for Omics Imaging

June 2024- May 2025

Advisor: Dr. Nils Gehlenborg

Harvard Medical School

- Developed interactive ipywidgets to visualize large-scale spatial omics datasets and built a chat-based interface enabling natural language queries in jupyter notebook.
- Integrated few-shot learning and retrieval-augmented generation (RAG) to improve information retrieval accuracy, structured outputs, and overall system reliability
- Contributed 5K+ lines of production-level Python code.

Undergraduate Research Assistant - SSL for Medical Imaging

May 2023 - May 2024

Advisor: Dr. Jacob Lubner

Health Data Science Lab

- Curated gigabyte-scale datasets ranging from histopathology to highly-plexed omics imaging.
- Created a similarity search framework that retrieved the most similar spatial proteomics slide based on input histopathology slide for in-depth biomarker exploration.
- Trained deep learning frameworks (VAE) on A100 NVIDIA GPUs for large-scale pathology and spatial proteomics datasets to develop the search framework.

Undergraduate Research Assistant - Hybrid Design and Fabrication

Jan 2023 - May 2024

Advisor: Dr. Cesar Torres

Hybrid Atelier

- Lasercut and 3D-Printed molds for developing silicone samples with different micro-textures to enhance haptic perception and study the effect on emotions and decision making.
- Created a web application using React and Flask API to collect user responses for study and observe texture perception of silicone samples.
- Developed a layered silicone composite sensor with unique electrode designs to study new gesture creation and cross-modal correspondences.

- Achieved 97% accuracy by training a CNN model to predict user gestures based on sensor reading heatmaps.

Undergraduate Research Assistant - Cognitive Science

Aug 2021- June 2022

Advisor: Dr. Matthew Robison and Dr. Pedro Maia

Controlled Attention and Memory Lab

- Developed python code to provide useful insight on average reaction time and pupil diameter/gaze shift obtained from eye-tracking devices for different conditions for 300+ participants.
- Calculated the sample entropy of pupil diameter to determine the regularity of pupil diameter based on type of goal, feedback, and reward.

PUBLICATIONS

[1] Amir Hajighasemi*, M. D. Saurav*, **Aarti Darji***, Mohammad S. Nasr, Jai Prakash Veerla, Parisa Boodaghi Malidarreh, Michael Robben, Helen H. Shang, and Jacob M. Luber. "Multimodal Pathology Image Search Between H&E Slides and Multiplexed Immunofluorescent Images." 2024 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB). IEEE, 2024. (Short Paper)

[2] **Darji, Aarti**, Gunnika Kapoor, and Cesar Torres. 'Trace to Touch: Eliciting Gestures from Capacitive Touch Electrodes.' Proceedings of the 16th Conference on Creativity & Cognition 2024. (Extended Abstract – Short Paper)

[3] Kapoor, Gunnika, **Aarti Darji**, Marisa N. Fernandez, and Cesar Torres. 'Sentura: Understanding the Cognitive Affordances of Silicone Microtextures in Tangible User Interface Design.' In Proceedings of the 16th Conference on Creativity & Cognition. (Extended Abstract – Short Paper)

[4] Veerla, Jai Prakash, Partha Sai Guttikonda, Amir Hajighasemi, Jillur Rahman Saurav, **Aarti Darji**, Cody T. Reynolds, Mohamed Mohamed, Mohammad S. Nasr, Helen H. Shang, and Jacob M. Luber. SpatialVisVR: An Immersive, Multiplexed Medical Image Viewer With Contextual Similar-Patient Search. 2024 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB). IEEE, 2024.

POSTER & PRESENTATIONS

[1] **Darji, Aarti***, Amir Hajighasemi*, M. D. Saurav*, Mohammad S. Nasr, Jai Prakash Veerla, Parisa Boodaghi Malidarreh, Michael Robben, Helen H. Shang, and Jacob M. Luber. *Integrated Search Framework for Histopathology: From H&E to Multiplexed Immunofluorescence*. Intelligent Systems of Molecular Biology (ISMB) 2024 (Poster).

[2] **Aarti Darji**, Eric Moerth, Morgan Turner, David Kouril, Jacob Luber, Nils Gehlenborg. *LLM-Supported Exploration and Analysis of 3D-Microscopy Imaging*. Bio+MedVis Challenge @ IEEE VIS 2024 (Talk & Poster).

[3] **Aarti Darji**, Eric Mört, Morgan Turner, David Kouřil, Nils Gehlenborg. *Ask and Reveal: Natural Language-Driven Analysis of Multiplexed Immunofluorescence Imaging*. Intelligent Systems of Molecular Biology (ISMB) 2025 (Poster).

OTHER PUBLICATIONS

[1] **Darji, Aarti**. "Revolutionizing Cancer Research with AI: Health Data Science Lab, UT Arlington." *XRDS: Crossroads, The ACM Magazine for Students* 30, no. 4 (2024): 64-65.

AWARDS & SCHOLARSHIPS

Maverick Academic Scholarship 2021-2025, College of Science Dean's Leadership Circle Award 2022-2023 & 2023-2024, OurCS@DFW 2023 Best Presentation Award, TExBioMed Summer Program 2023 Outstanding Presentation Award, Honors College Dean Scholarship 2024-2025

LEADERSHIP

Vice President, *Psychology Society*

Dec 2022 – May 2023

Research Director, *ACM@UTA*

Sep 2022 – May 2023

Programming Board Coordinator, *Honors College*

Sep 2022 – May 2023

Veneratio Newsletter Staff, *Honors College*

Sep 2022 - Dec 2022